

DRY-RUN GUIDE | dojo2.py

Source: Python\2nd_Belt\dojo2.py

CORE BEHAVIOR

Validates a non-negative integer string and repeatedly sums digits until one digit remains.

REPRESENTATIVE INPUT

'9875'

STEP TRACE

9+8+7+5=29; then 2+9=11; then 1+1=2; stop because 2<10.

EXPECTED OUTPUT / RETURN

'Single-digit sum: 2'

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\2nd_Belt\dojo2.py; line numbers refer to the original file.
No source was modified.

```
001 | # Question: Solve the given Dojo challenge using the specified logic.
```

```
002 | # Add your solution here.
```

```
003 |
```

```
004 | def sum_digits_until_single(num):
```

```
005 |
```

```
006 |     while num >= 10:
```

```
007 |         total = 0
```

```
008 |         while num > 0:
```

```
009 |             total += num % 10
```

```
010 |             num //= 10
```

```
011 |         num = total
```

```
012 |     return num
```

```
013 |
```

```
014 | user_input = input("Enter a non-negative integer: ").strip()
```

```
015 |
```

```
016 | if not user_input.isdigit():
```

```
017 |     raise ValueError("Input must be a non-negative integer.")
```

```
018 |
```

```
019 | number = int(user_input)
```

```
020 |
```

```
021 | result = sum_digits_until_single(number)
```

```
022 | print(f"Single-digit sum: {result}")
```

```
023 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.